

Introductory Calculus



Direct substitution

- 1 Find $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$.
 - 2 Find $\lim_{x \rightarrow -1} (x^3 - 5x)$.
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Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.
 - 4 Find $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$.
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The derivative as a limit

- 5 Write the difference-quotient definition of $f'(x)$.
 - 6 Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 - 7 Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.
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Differential equations

- 8 Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.
- 9 Explain why solving a differential equation gives a family of functions rather than a single number.